

Stage 1

Multi-Signal Exposure Mapping



S1: Embedding Similarity



S2: Min-K%++



S4: Self-Critique

$\text{RRF} \rightarrow e_{ij} \in [0, 1]$

Reciprocal Rank Fusion

e_{ij}

Stage 2

Contamination-Augmented IRT

$$P(y_{ijk} = 1) = \sigma(a_i(\theta_j - b_i) + \gamma_k e_{ij})$$

Bayesian NUTS (4 chains)

MCMC Inference

$\theta_j^{\text{clean}}, \gamma_k + \text{CIs}$

Decontaminated estimates

θ_j, γ_k

Stage 3

Dose-Response Profiling

Algorithm 1 classification

Immune

Collapse

V-recovery

Benefit